

PhD researcher in solar physics, developing analytic and MHD-simulation models of magnetic switchbacks in the solar wind, supervised by Dr Peter Wyper and Prof. Anthony Yeates.

Combines this with undergraduate teaching and sustained public engagement in mathematics communication.

Research experience

Doctor of Philosophy (PhD) in Applied Mathematics

Oct 2025 – Present

Durham University · Supervisors: Dr Peter Wyper & Prof. Anthony Yeates

- Developing 3D analytic and numerical models of magnetic switchbacks: rapid, localised reversals in the Sun's ambient magnetic field that would otherwise be radial.
- Extending existing analytic treatments of switchback structure and theory on spherically-polarised Alfvén waves; validating predictions against MHD simulations run in ARMS and Lare3d.

Integrated Master of Mathematics (MMath)

Sept 2021 – July 2025

Durham University · First Class Honours · Supervisor: Dr Madalena Lemos

- **Dissertation:** *The Standard Model from a symmetry perspective.*
- Researched the construction of the Standard Model of Particle Physics via gauge theory and spontaneous symmetry breaking; produced a written report, an academic poster and an oral presentation ([available on personal website](#)).

Presentations, posters & talks

- **Magnetic switchbacks and how to model them** — **Invited talk**, SSXG Solar Science Meeting, Center for Astrophysics | Harvard & Smithsonian, September 2026
- **Magnetic switchbacks and how to model them** — Internal talk, [Durham Applied Maths seminar](#), July 2026
- **Switchbacks 101** — ‘Lightning’ presentation and poster, workshop for the [North East Group for the Mathematics of Complexity and Multi-Scale Phenomena](#), Northumbria University, June 2026
- **Analytic, axisymmetric solutions for magnetic switchbacks** — **Awarded best student poster**, UKMHD 2026, University of Glasgow, June 2026

Teaching experience

Undergraduate tutorial leader

Oct 2025 – Present

Durham University

- Teaching assistant for Fluid Mechanics III; tutorial leader for Mathematical Methods II and Methods of Mathematical Physics II.
- Guide small groups through problem sheets, providing mentorship alongside teaching.

Maths and science tutor

Remote

[MyTutor.co.uk](#) (2022 – 2025) · *Freelance* (2025 – Present)

- Delivered one-to-one and small-group tuition via video call; sustained 100% return business rate on the MyTutor platform.
- Now private tutor on a word-of-mouth basis.

Outreach programs

PANDURS

Since Sept 2026

Event co-lead

- Co-organising the university's [latest seminar series](#): Postgraduate **AND** Undergraduate **R**esearch **S**eminar.
- Hosting lectures, coordinating career panels and skills workshops, encouraging discussion.

24 Hours of Le Maths

Since Dec 2025

Lead event organiser · Durham University Mathematical Problem Solving Society

- Organising the university's annual [24-hour maths endurance event](#); managing a team of volunteer question writers and markers and liaising directly with the beneficiary charity, [Point North](#).

Chalkdust magazine

Since Feb 2024

Article writer and editor · Cryptic crossword setter (as "Seuss")

- Writing and editing articles for a popular maths magazine, distributed among universities internationally.
- [Most recent issue](#).

Chalkboard Ultra podcast

Oct 2023 – June 2025

Co-host and lead editor

- Hosted conversations with PhD students and lecturers across all areas of mathematics. [Spotify playlist](#).

Skills

Computational	Python, ARMS, LareXd, Maple, VisualPDE, $\text{\LaTeX}$, TikZ.
Communication	Technical writing & editing, public-facing science communication, presentation and poster design.
Miscellaneous	Microsoft Excel, sleight of hand and card magic.

Additional activities

- Maths subject representative for Stephenson College since 2025: advisory point of contact for exam technique, module choices and academic issues.
- Summer camp counsellor and group leader at Island Lake Camp in Starrucca, PA, USA.
- Worked remotely as a data research assistant and analyst, supporting the creation, deployment and analysis of a study with Cranfield University on the impacts of sales coaching.
- Jazz music enthusiast; have played guitar, keyboard & keytar for various bands over the years.

References

Dr Peter Wyper

Associate professor · Durham University

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